TECHNICAL SPECIFICATIONS

NUCLEAR DATA INC.
100 West Golf Road
Palatine, Illinois 60067
Phone A.C. 312 • 529-4600

SERIES 3300 SYSTEM MEMORY Unit

The System Memory Unit is a coincident current magnetic core type. Its function is to store data supplied by System Data Acquisition Units, or other external sources, and to supply this data for either further processing or readout upon command. Storage capacity can be either 4096 twenty-bit words, 16,384 eighteen-bit words, or 16,384 twenty-bit words, as specified when ordered. The time required to complete a full read-add one-write storage cycle for one word of data is four microseconds.

Included in the System Memory Unit are a magnetic core stack, memory drive circuitry, and associated address selection circuitry. Straight binary address and arithmetic coding are used to achieve the most efficient utilization of the core stack and associated binary circuits.

Number of Words (Channels)

The memory can store up to 4096, 8192, or 16,384 words of binary data, as specified when ordered.

Word Length

4096 -- Up to twenty binary bits of information can be stored at each word location. This provides a maximum capacity of 1,048,575 counts for totalization of data.

16,384 -- Either eighteen or twenty binary bits of information can be stored at each word location, as specified when ordered. This provides a maximum capacity of 262,143 or 1,048,575 counts for totalization of data.

Storage Cycle Time

One word of data is extracted from the memory, modified, and then re-inserted within four (4) microseconds (Read, Add-1, Write).

Sense Amplifiers and Strobe

4096 -- The memory contains twenty sense amplifiers and strobe circuitry used in reading information from the magnetic core stack.

16,384 -- The memory contains either eighteen or twenty sense amplifiers and strobe circuitry used in reading information from the magnetic core stack, as specified when ordered.

Inhibit Circuits

4096 -- The memory contains twenty inhibit circuits used in writing information into the magnetic core stack.

16,384 -- The memory contains either eighteen or twenty inhibit circuits used in writing information into the magnetic core stack, as specified when ordered.

Word Selection

The memory contains the word selection circuitry used in writing information into and reading information out of the magnetic core stack.

Data Retention

No more than one word of data will be dropped if a power failure occurs, or if the power is turned off when the system is not in the STOP mode.

Temperature Range

The memory will operate satisfactorily within the range of 15° to 35° Centigrade.

Size

14" high x 19" wide x 22" deep.

Weight

35 pounds.